

||Jai Sri Gurudev ||
BGSKH Education Trust (R.) – A unit of Sri Adichunchanagiri Shikshana Trust(R.)

BGS College of Engineering and Technology (BGSCET)
Mahalakshmipuram, West of Chord Road, Bengaluru-560086
(Approved by AICTE, New Delhi and Affiliated to VTU, Belagavi)



BGS COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING – MAJOR PROJECT

TITLE: KNACK – A Gamified Platform for Skill Development

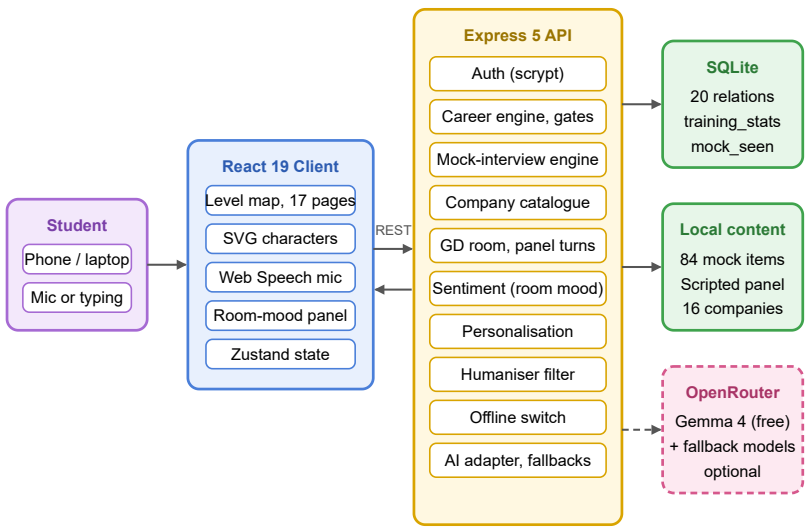
ABSTRACT

Knack is a gamified web platform that prepares engineering students for campus placements by simulating the whole journey: a group discussion, five interview rounds, joining, an internship, an offer and full-time work. Before every real round the student takes an MCQ mock in which every option is explained. Group discussions run with five profiled AI panellists who speak one at a time, beside a live, lexicon-based sentiment reading of the room's mood. Sixteen company profiles carry turnover, headcount, hiring steps and open roles. Every attempt is stored to personalise the next session, answers can be spoken, and a full offline mode keeps it working without the internet.

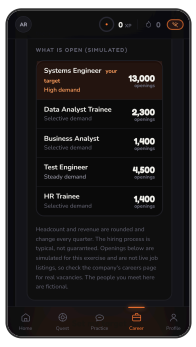
OBJECTIVE

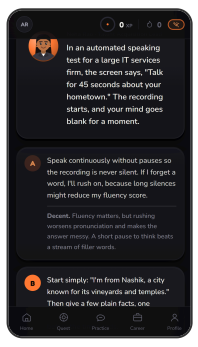
- To simulate the complete placement path, from group discussion to full-time work, with each stage unlocked only on demonstrated competency.
- To teach before testing through an MCQ mock before every interview round, with every option graded and explained.
- To make group-discussion practice realistic with profiled AI panellists and a live reading of the room's mood.
- To personalise practice from stored history, and keep the platform usable offline, by voice or by typing.

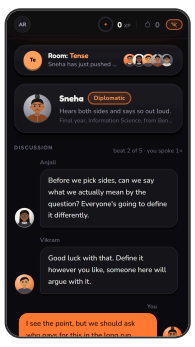
METHODOLOGY / SYSTEM ARCHITECTURE

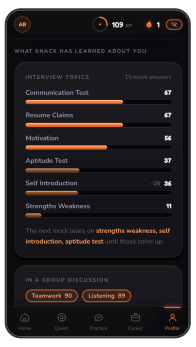


RESULTS & DISCUSSION


Company profile


Mock, explained


Room mood


Training profile

- 37 of 37 end-to-end checks passed in offline mode; median server response about 1 ms.
- Question selection wastes 41% fewer repeats than a no-repeat cycle (2000 simulated learners).

CONCLUSION

- Knack delivers a working, gamified simulation of the full placement journey, tied to real company profiles.
- Mock-first teaching shows students what a strong answer looks like before they are assessed.
- Profiled panellists and the room-mood reading make solo group-discussion practice realistic.
- Stored, visible personalisation and a full offline mode make practice continuous and always available.

FUTURE SCOPE

- A pilot study with students, rated before and after by independent assessors.
- Spoken replies from the interviewer and the panel, and AI-based sentiment analysis of the discussion.
- A larger question bank and company catalogue, and a native mobile app.

HW / SW REQUIREMENTS

Hardware:

- Laptop / PC: dual-core 2 GHz, 4 GB RAM, 500 MB storage
- Microphone (optional, for spoken answers)
- Any phone or laptop with a modern browser

Software:

- Node.js 24, Express 5, SQLite (better-sqlite3)
- React 19, Vite, Tailwind CSS, Framer Motion
- OpenRouter – Gemma 4 (optional AI)
- Web Speech API (Chrome / Edge)

STUDENT DETAILS / GUIDED

Presented By:
D Karthik Raj
Kiranraj G
Mohammed Ishaaq
Ullas M

Guided By:
Dr. Chaitra Naveen,
Dept. of ISE, BGSCET